

Chapter 13:

Feature-Based Flow Visualization

Prof. Dr.-Ing. Thomas Schultz

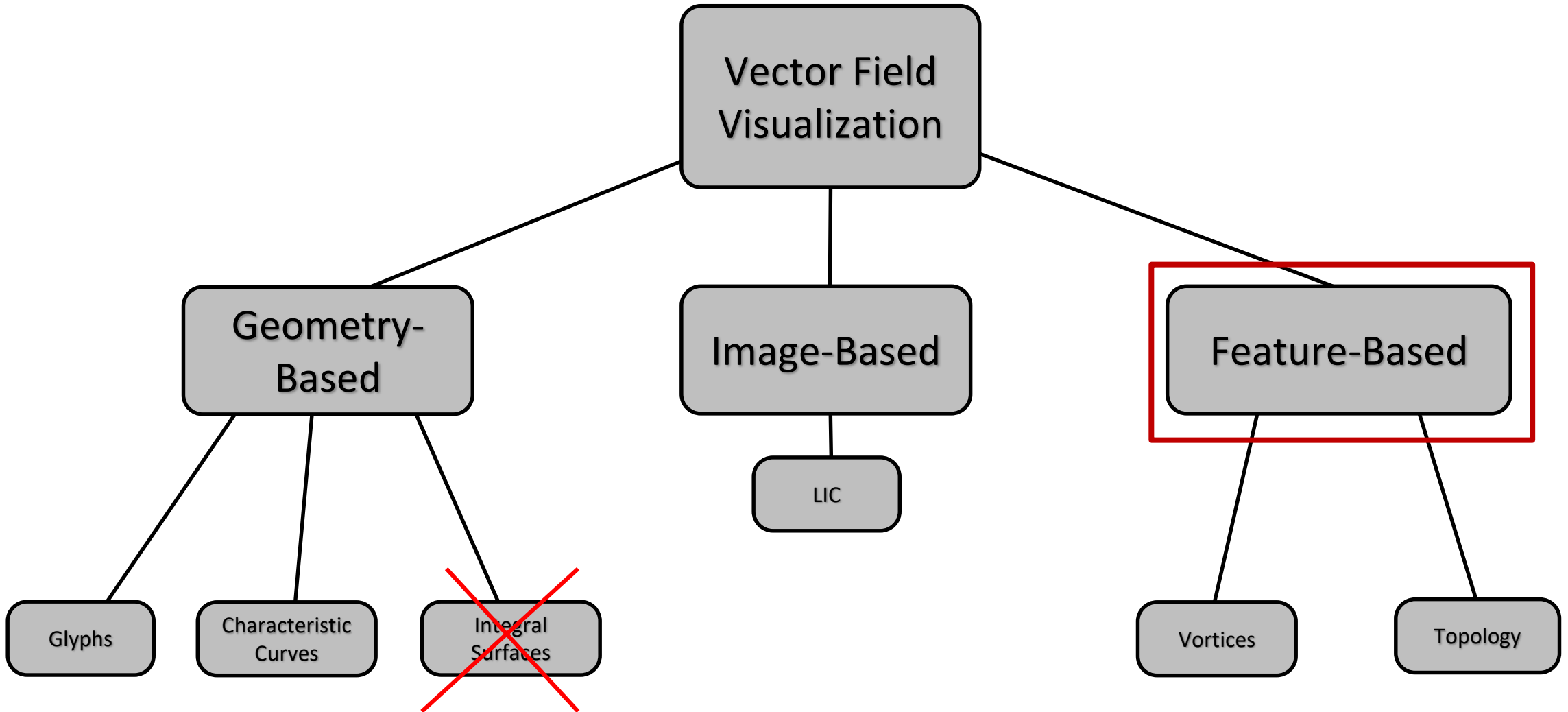
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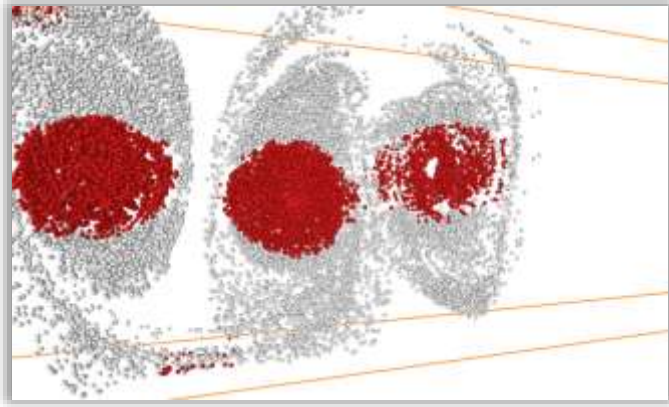
July 1/7, 2026

Overview of Vector Field Visualization



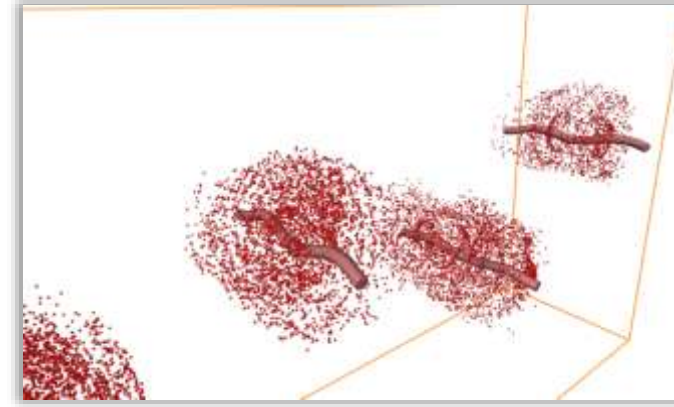
Classic vs. Feature-based Visualization

Classic Visualization



- Showing a feature
 - Human understands what the feature is
- Path? – Velocity? – Life time? – Interdependency? – Importance?
➔ Qualitative analysis

Feature-based Visualization



- Defining a feature and showing it
 - Human and Computer understand what the feature is
- Path! – Velocity! – Life time! – Interdependency! – Importance!
➔ Quantitative analysis (e.g. statistics)
➔ Qualitative analysis

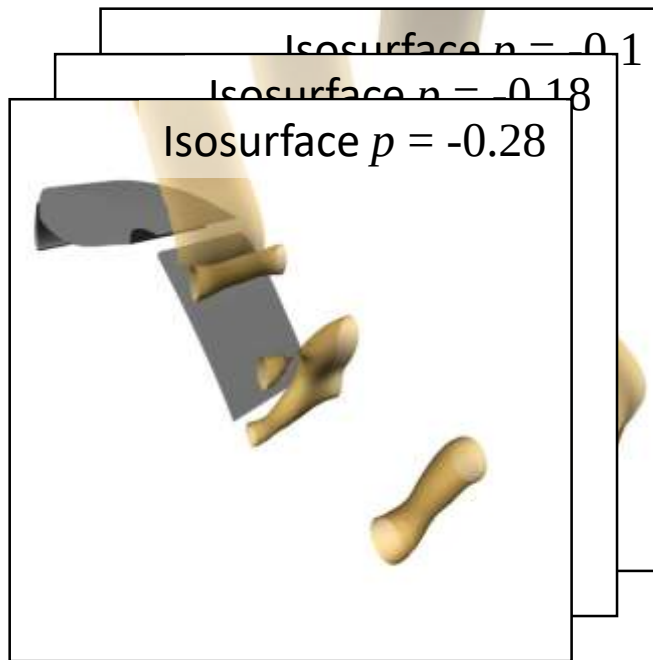
Feature-based Visualization

- **Goals and advantages**
 - Represent interesting structures or objects in the data
 - Formal mathematical definition of “interesting” structures
 - Algorithms for extraction and visualization
 - Extraction can (usually) be automated
 - The original (large) field is not needed anymore for visualization
 - Rendering is fast and (usually) interactive

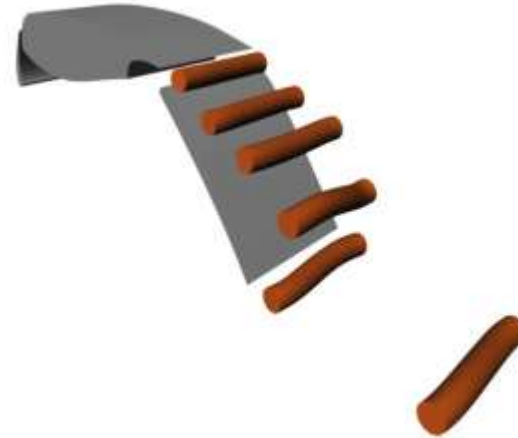
Example: Classic vs. Feature-based Visualization

- *Example:* Detection of vortex regions

Classic Visualization



Feature-based Visualization

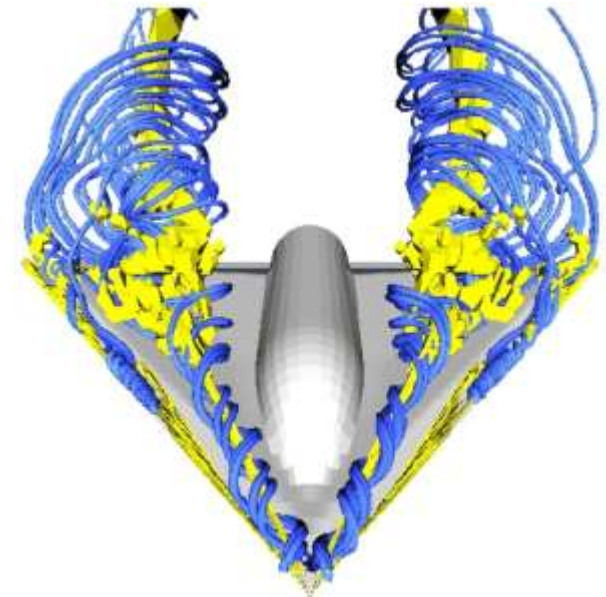
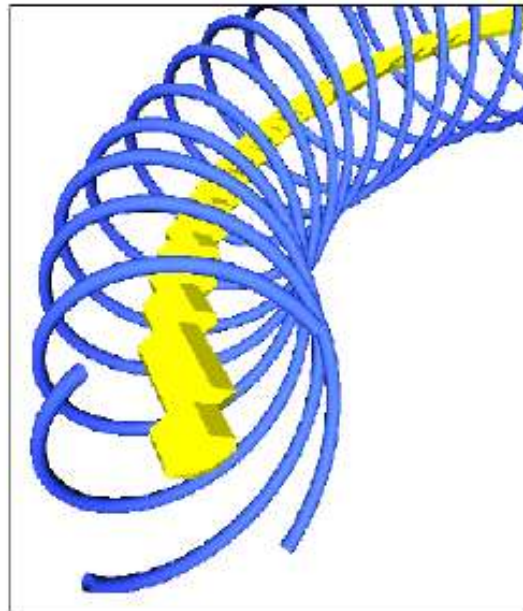
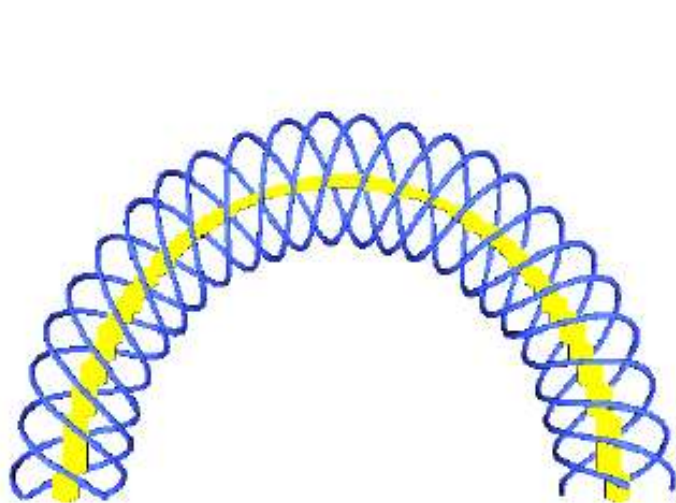


Section 13.1: Vortex Core Lines

Vortex Cores

Intuitively, a vortex is a region of swirling flow behavior (see Section 11.5)

- There is no universally accepted formal definition of what constitutes a vortex
- However, most definitions involve the idea of a „vortex core“ around which the swirling motion takes place

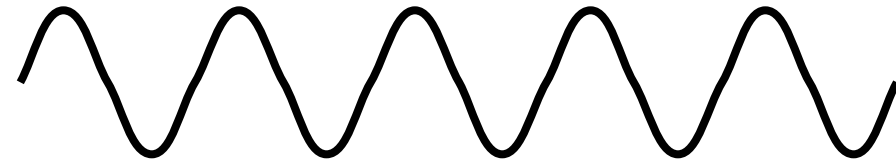
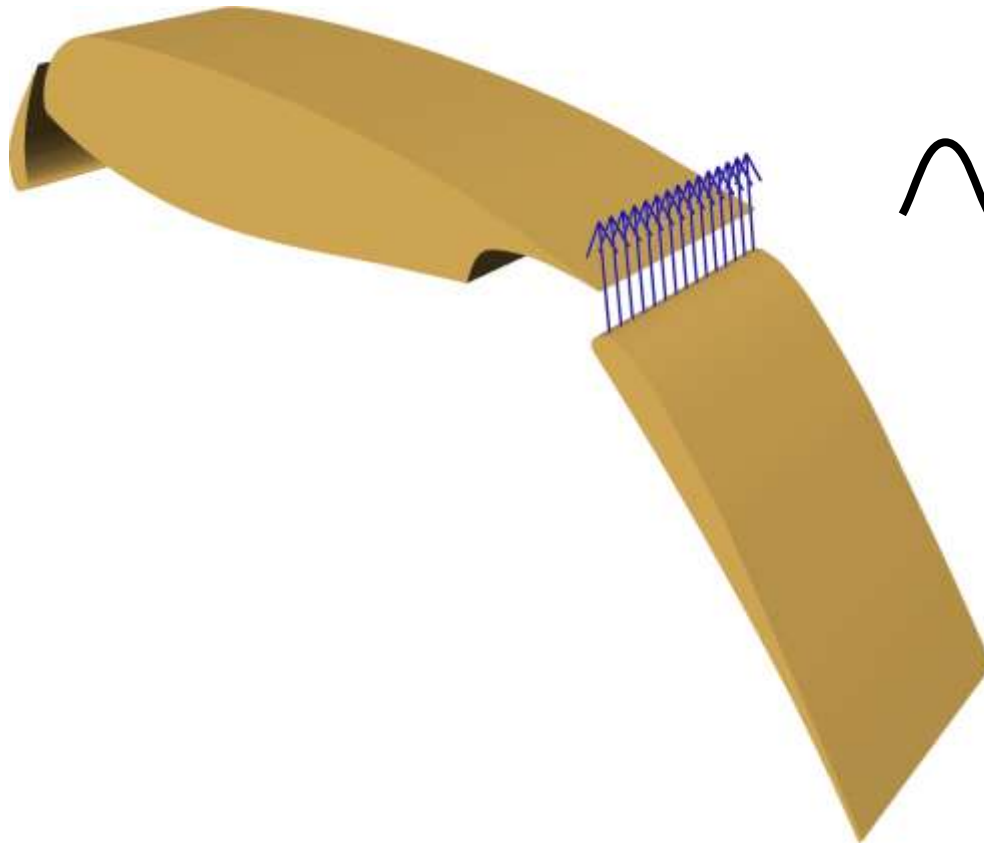


What is a Vortex? Some Definitions

- [Lugt 1972]
 - A vortex is the rotating motion of a multitude of material particles *around a common center*.
- [Portela 1997]
 - A vortex is comprised by *a central core region* surrounded by swirling stream lines.
- [Robinson 1991]
 - A vortex exists when instantaneous streamlines mapped onto a plane *normal to the vortex core* exhibit a roughly circular or spiral pattern, when viewed from a reference frame moving with the center of the vortex core.

Example: Relevance of Vortices

Numerical Experiment: Optimize lift of a wing by periodic blowing and suction at the rear flap

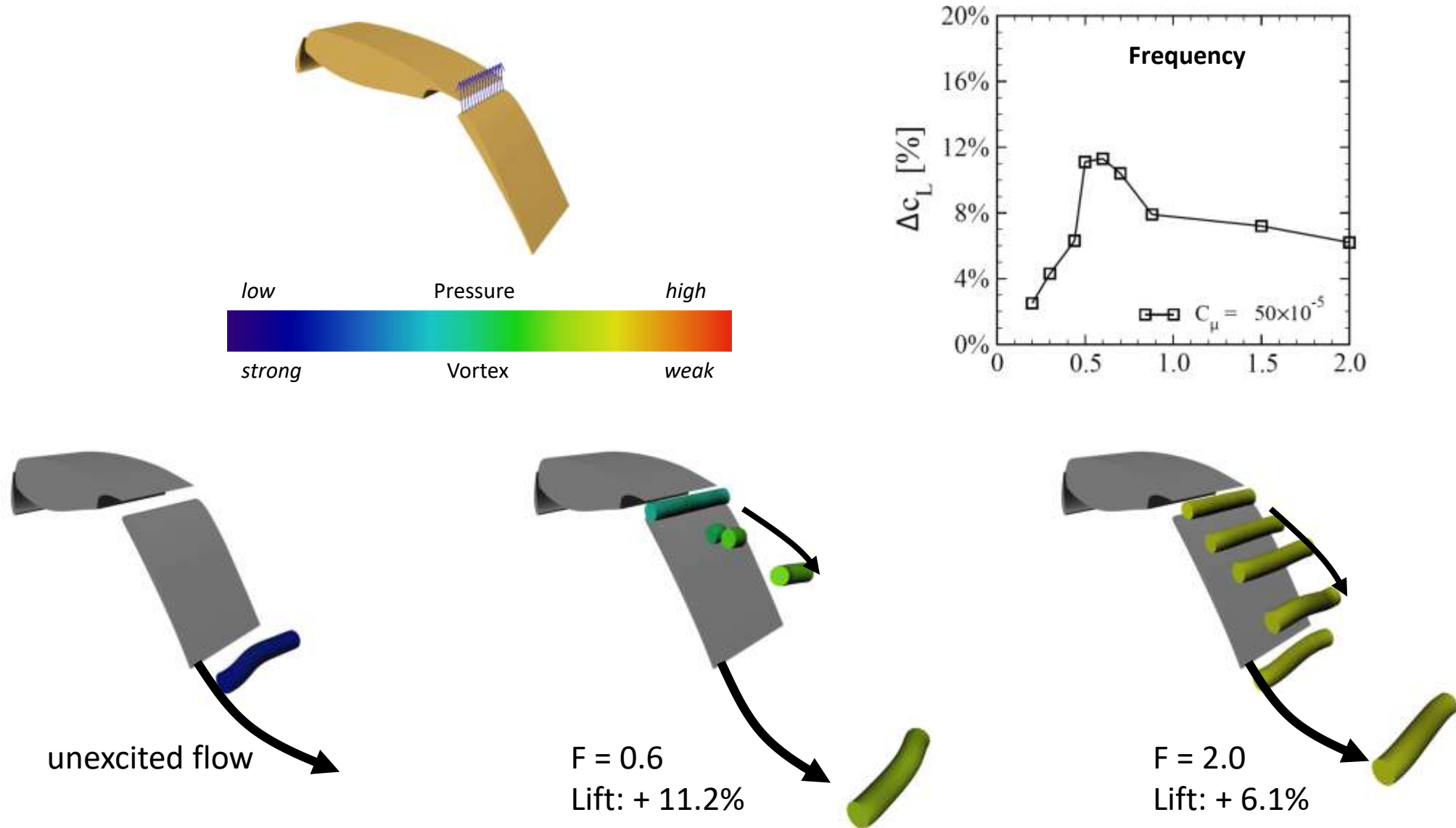


Parameters:

Frequency of air injection

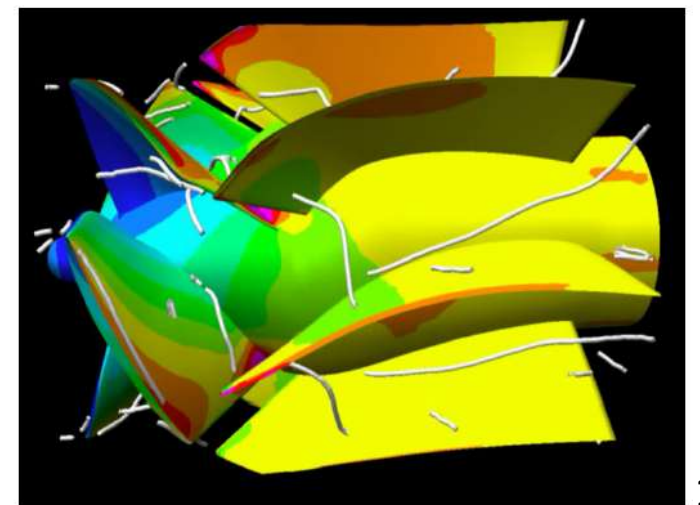
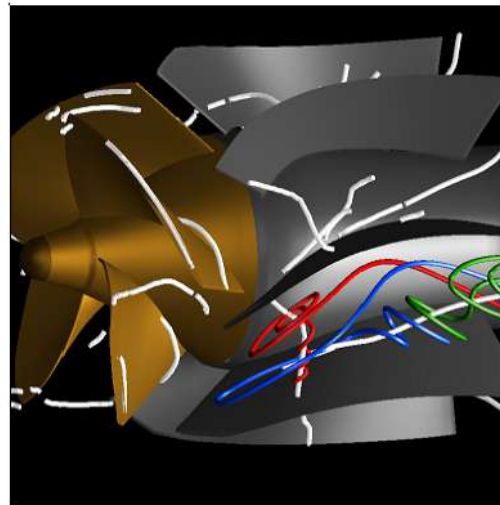
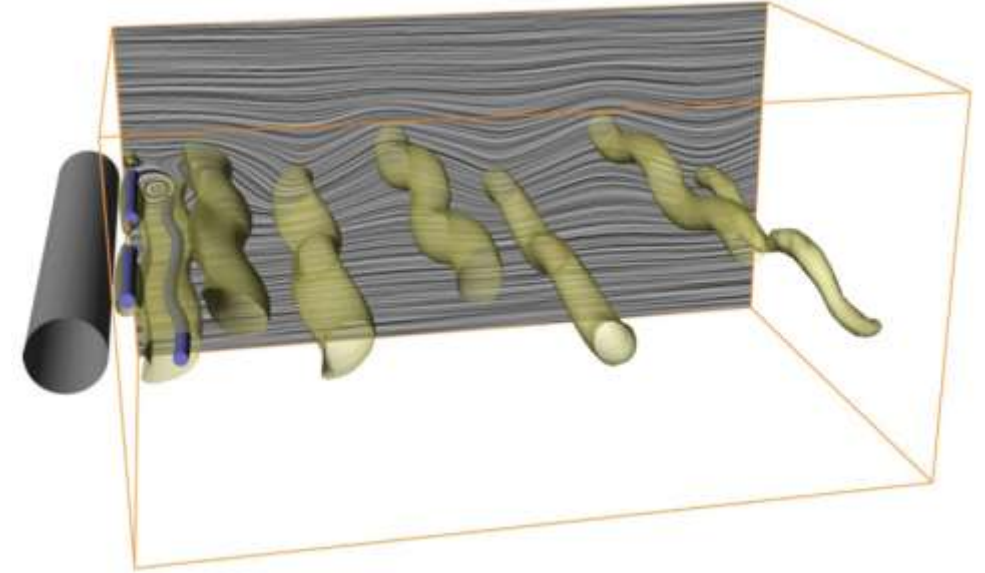
Intensity of air injection

Example: Relevance of Vortices



Classification of Vortex Visualizations

- **Two main classes of vortex structures**
 - **Region-based:** visualize a region of swirling flow behavior (e.g., based on vorticity/ Q , as defined in Section 11.5)
 - **Line-based:** extract vortex core lines

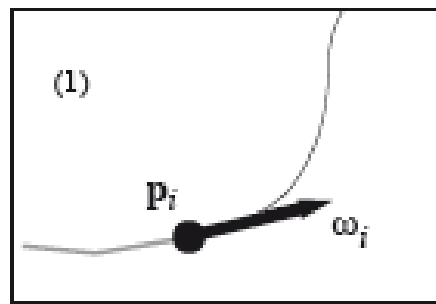


Prediction-Correction Scheme

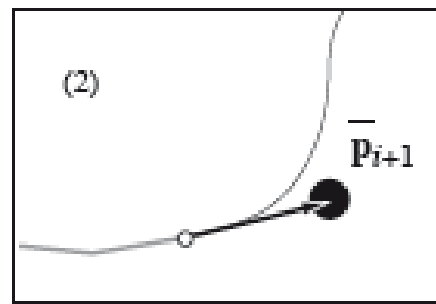
Idea of [Banks/Singer 1994]:

Extract lines that should, at the same time, approximately follow the direction of vorticity and coincide with local minima of pressure

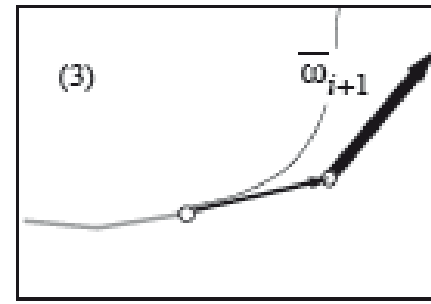
- **Predict:** Take a step in direction of vorticity
- **Correct:** To a pressure minimum in the perpendicular plane



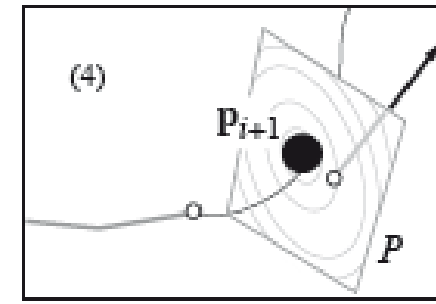
Compute the vorticity at a point on the vortex core.



Step in the vorticity direction to predict the next point.



Compute the vorticity at the predicted point.



Correct to the pressure min in the perpendicular plane.

Details on Prediction-Correction

- Requires **seed points**
 - e.g., start at local minima of pressure
 - Need to identify duplicates that might be caused by multiple seed points on the same core
- Requires an iterative **gradient descent** in the correction step
- Requires **stopping criterion**
 - e.g., stop when correction step is too large

Parallel Vectors Operator

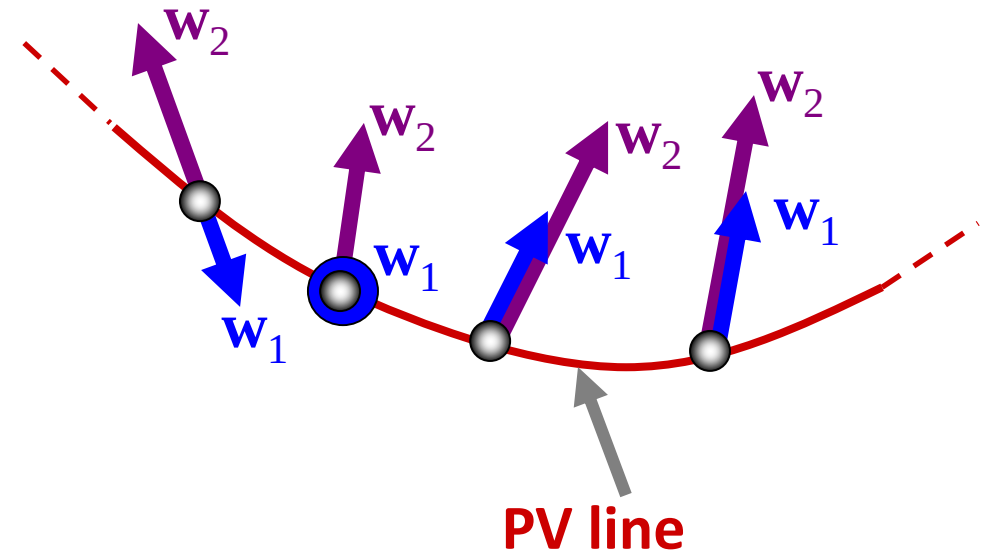
PV = Parallel Vectors [Peikert/Roth 1999]

Given are two 3D vector fields $\mathbf{w}_1, \mathbf{w}_2$.

The locations at which $\mathbf{w}_1 \parallel \mathbf{w}_2$ form curves:

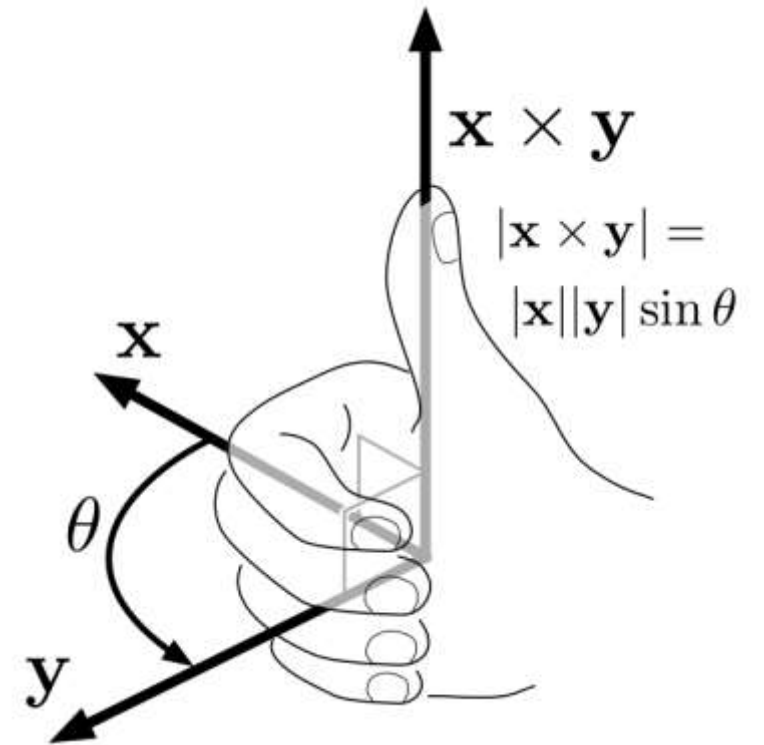
Alternative formulations:

- $\mathbf{w}_1 \parallel \mathbf{w}_2$
- $\mathbf{w}_1 = \lambda \mathbf{w}_2$ (for $-\infty \leq \lambda \leq \infty$)
- $\lambda_1 \mathbf{w}_1 + \lambda_2 \mathbf{w}_2 = 0$ (for $\sum \lambda_i^2 > 0$)
- $\mathbf{w}_1 \times \mathbf{w}_2 = (0,0,0)^T$



Reminder: Cross Product

- The **cross product** $\mathbf{x} \times \mathbf{y}$ of 3D vectors $\mathbf{x}$ and $\mathbf{y}$ is another vector whose
 - Direction is perpendicular to both $\mathbf{x}$ and $\mathbf{y}$
 - Length is $\|\mathbf{x} \times \mathbf{y}\| = \|\mathbf{x}\| \|\mathbf{y}\| \sin \theta$
 - Orientation is defined by the **right-hand rule**



Extracting PV Lines by Surface Intersection

- Based on $\mathbf{w}_1 \times \mathbf{w}_2 = (0,0,0)^T$
 - Extract zero isosurfaces for two coefficients of cross product
 - Find curves along which they intersect
 - Test if third coefficient is also zero

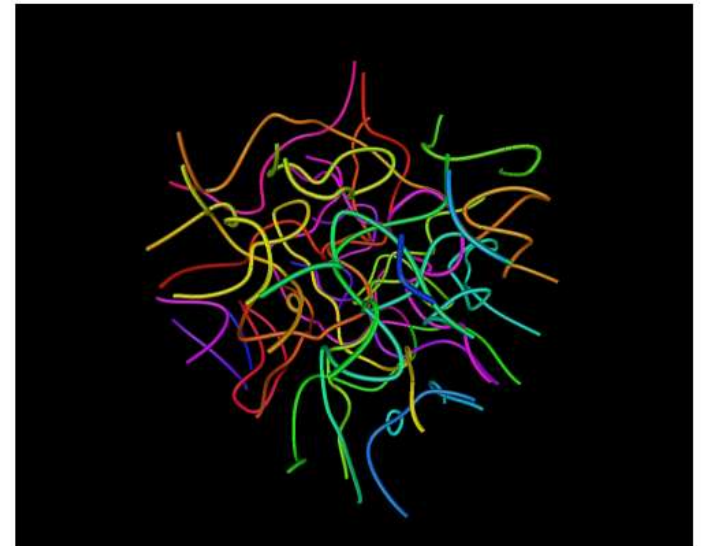
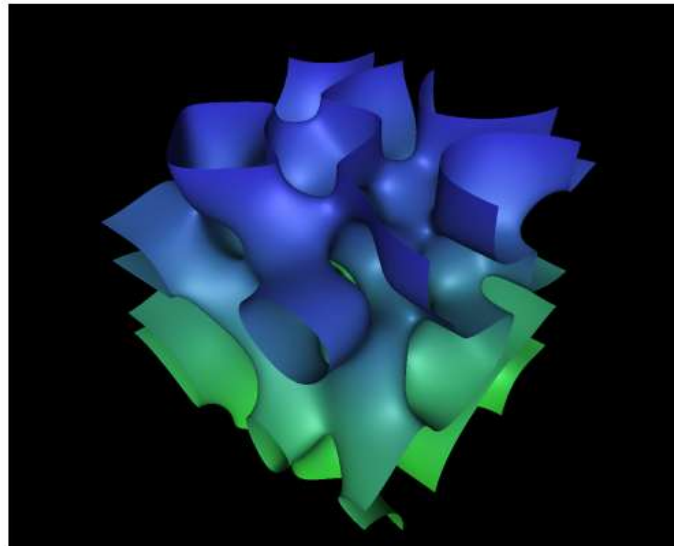
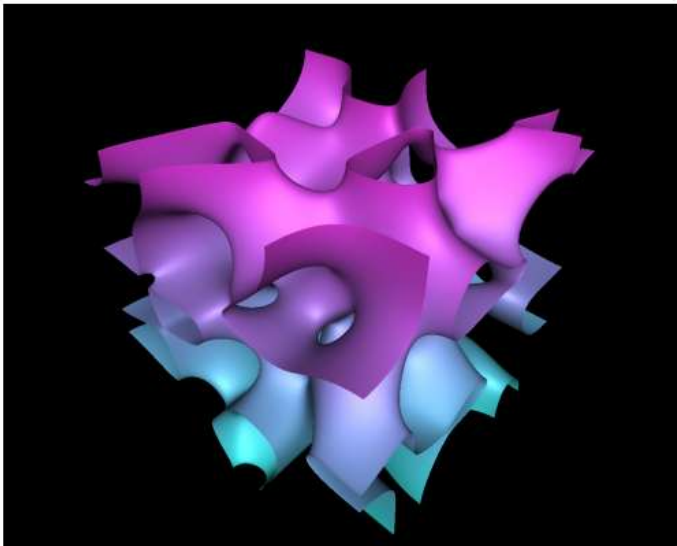


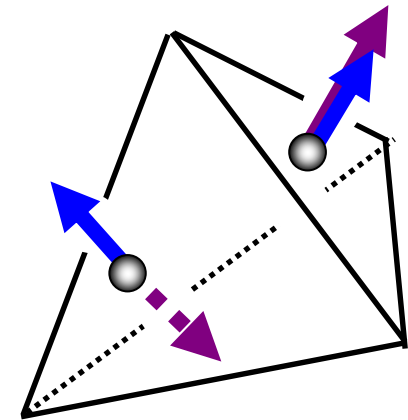
Illustration from [Ljung and Ynnerman, 2003]

Algebraic Localization of PV Lines

- Localize intersection with faces of tetrahedra, connect them
 - Based on $\mathbf{w}_1 = \lambda \mathbf{w}_2$ (for $-\infty \leq \lambda \leq \infty$)
 - Assuming linear interpolation, can be solved as an eigenvalue problem

$$- \mathbf{w}_1 = W_1 \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}, \mathbf{w}_2 = W_2 \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$

$$- W_1 \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \lambda W_2 \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} \Leftrightarrow W_2^{-1} W_1 \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix} = \lambda \begin{bmatrix} \alpha \\ \beta \\ \gamma \end{bmatrix}$$



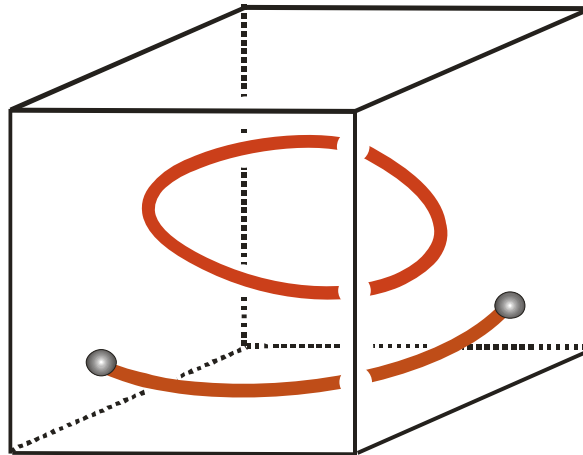
Properties of PV Lines

Property: PV lines are either closed or intersecting the boundary of the domain

- Intersection of closed isosurfaces
- Lines, because we are only restricting the 2D part

$$w_1 \perp w_2$$

- Vortex cores are *subsets* of these lines; they may not be closed!



Vortex Cores as PV Lines

We obtain necessary conditions for vortex cores through different choices of $\mathbf{w}_1, \mathbf{w}_2$:

- $\mathbf{w}_1 = \nabla \times \mathbf{v}$, $\mathbf{w}_2 = \nabla p$ [Banks/Singer 94] Similar idea as prediction / correction
- $\mathbf{w}_1 = \mathbf{v}$, $\mathbf{w}_2 = \nabla \mathbf{v} \mathbf{v}$ [Sujudi/Haimes 95] Jacobian eigenvector, see next slides
- $\mathbf{w}_1 = \mathbf{v}$, $\mathbf{w}_2 = \nabla(\nabla \mathbf{v} \mathbf{v}) \mathbf{v}$ [Roth/Peikert 98] Alternative for weak curved cores (no details)

Additional conditions lead to *filtering criteria* to find those PV lines (and segments) that correspond to vortex cores.

Swirling motion of stream lines – 3D steady case

Eigenvector method

[Sujudi and Haimes, AIAA 1995]

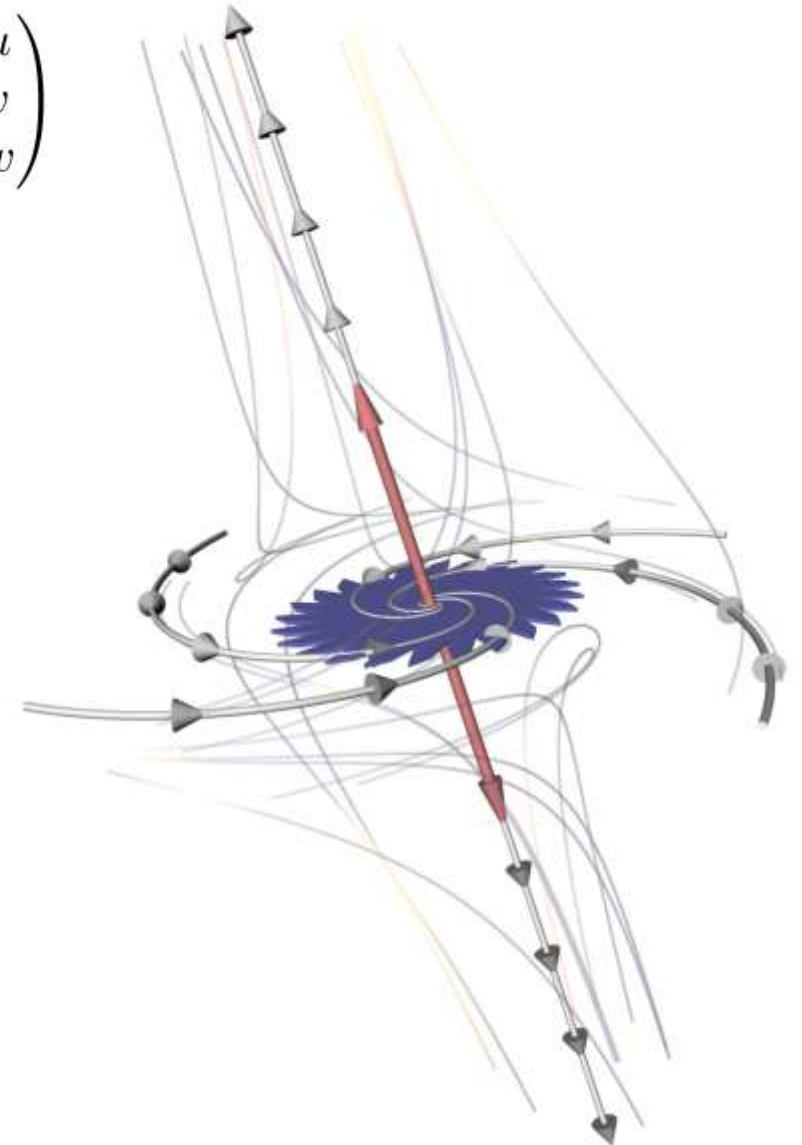
$$\mathbf{w}(\mathbf{x}) = \mathbf{v}(\mathbf{x}) - (\mathbf{v}(\mathbf{x}) \cdot \mathbf{e}(\mathbf{x})) \mathbf{e}(\mathbf{x})$$

$$\mathbf{w}(\mathbf{x}) = \mathbf{0}$$

reduced velocity

Although a point $\mathbf{x}$ on the core structure is surrounded by spiraling integral curves, the flow vector at $\mathbf{x}$ itself is solely governed by the non-swirling part of the flow.

$$\mathbf{v} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$



Swirling motion of stream lines – 3D steady case

Eigenvector method

[Sujudi and Haimes, AIAA 1995]

$$\mathbf{w}(\mathbf{x}) = \mathbf{v}(\mathbf{x}) - (\mathbf{v}(\mathbf{x}) \cdot \mathbf{e}(\mathbf{x})) \mathbf{e}(\mathbf{x})$$

$$\mathbf{w}(\mathbf{x}) = \mathbf{0}$$

reduced velocity

$$\mathbf{v} = \begin{pmatrix} u \\ v \\ w \end{pmatrix}$$

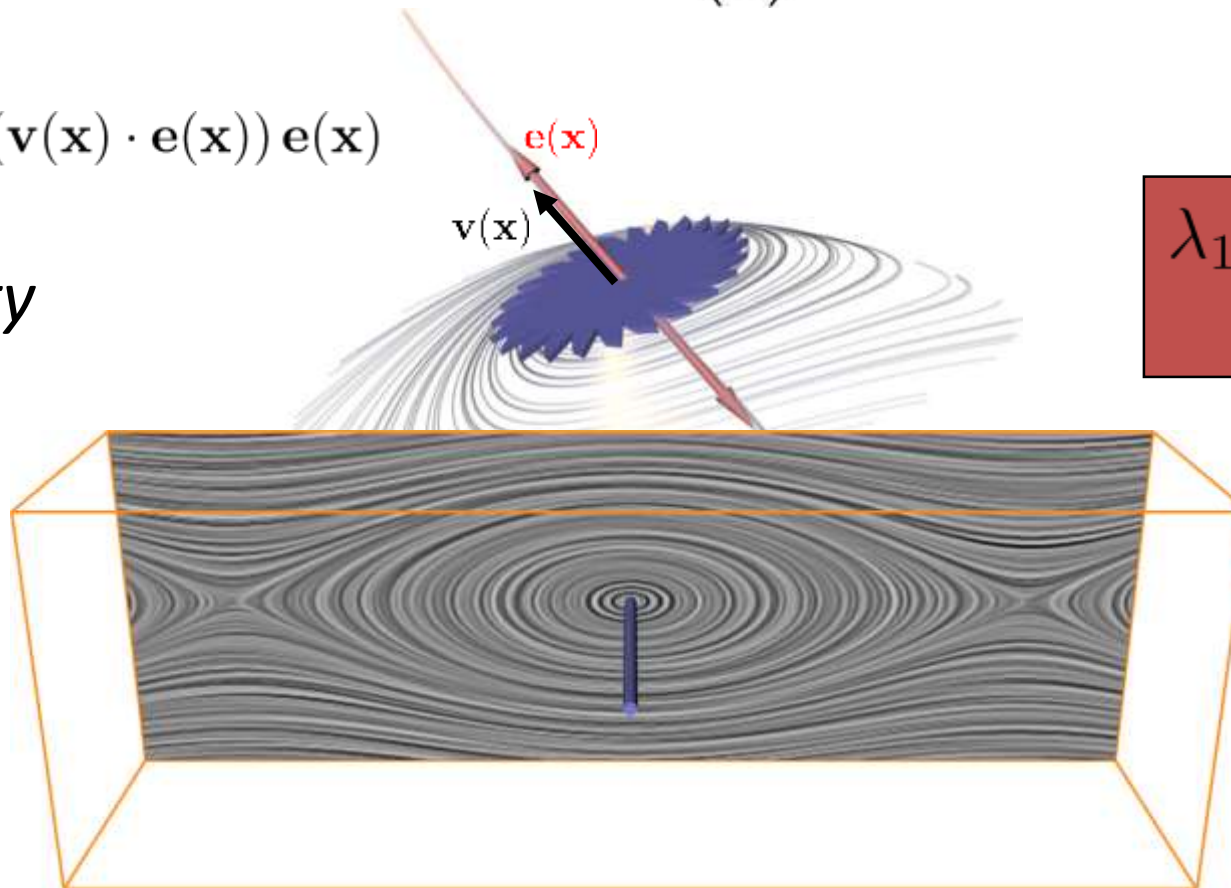
Parallel Vectors operator

[Peikert and Roth, Vis 1999]

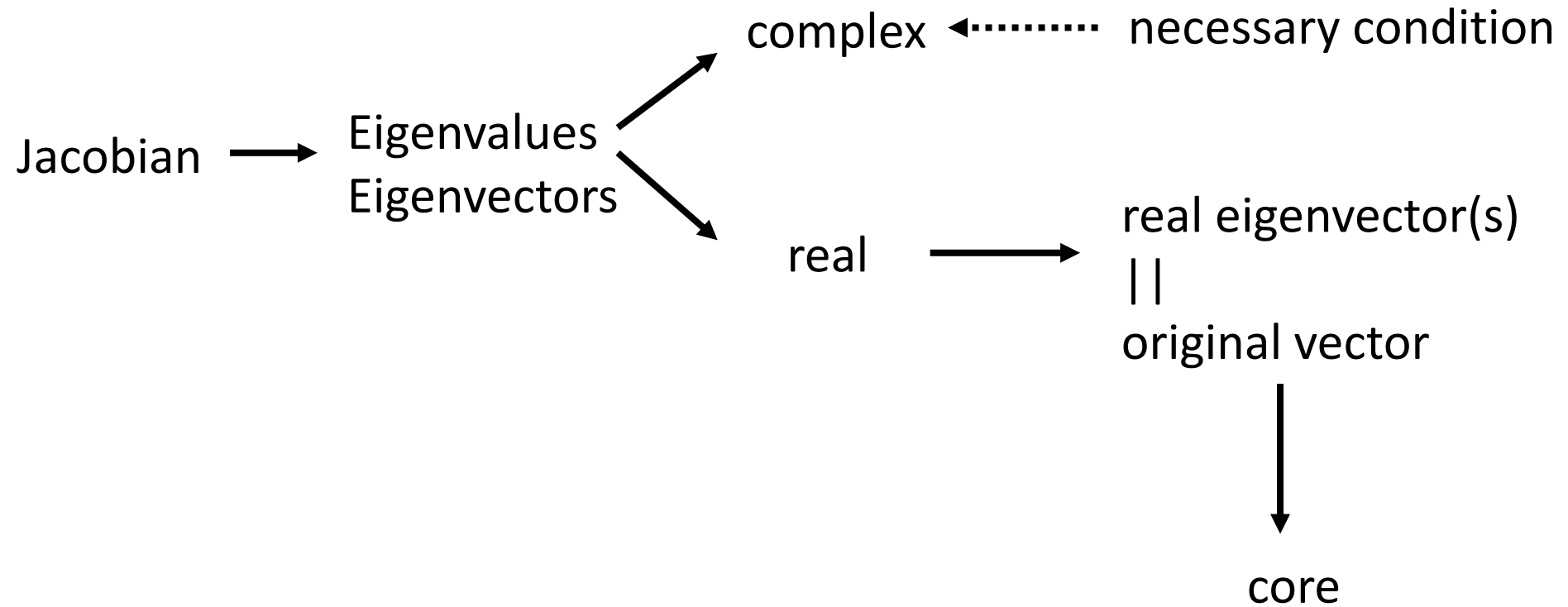
$$\mathbf{v}(\mathbf{x}) \parallel \mathbf{e}(\mathbf{x})$$

$$\lambda_1 \mathbf{v}(\mathbf{x}) + \lambda_2 \mathbf{e}(\mathbf{x}) = \mathbf{0}$$

with $\sum \lambda_i^2 > 0$

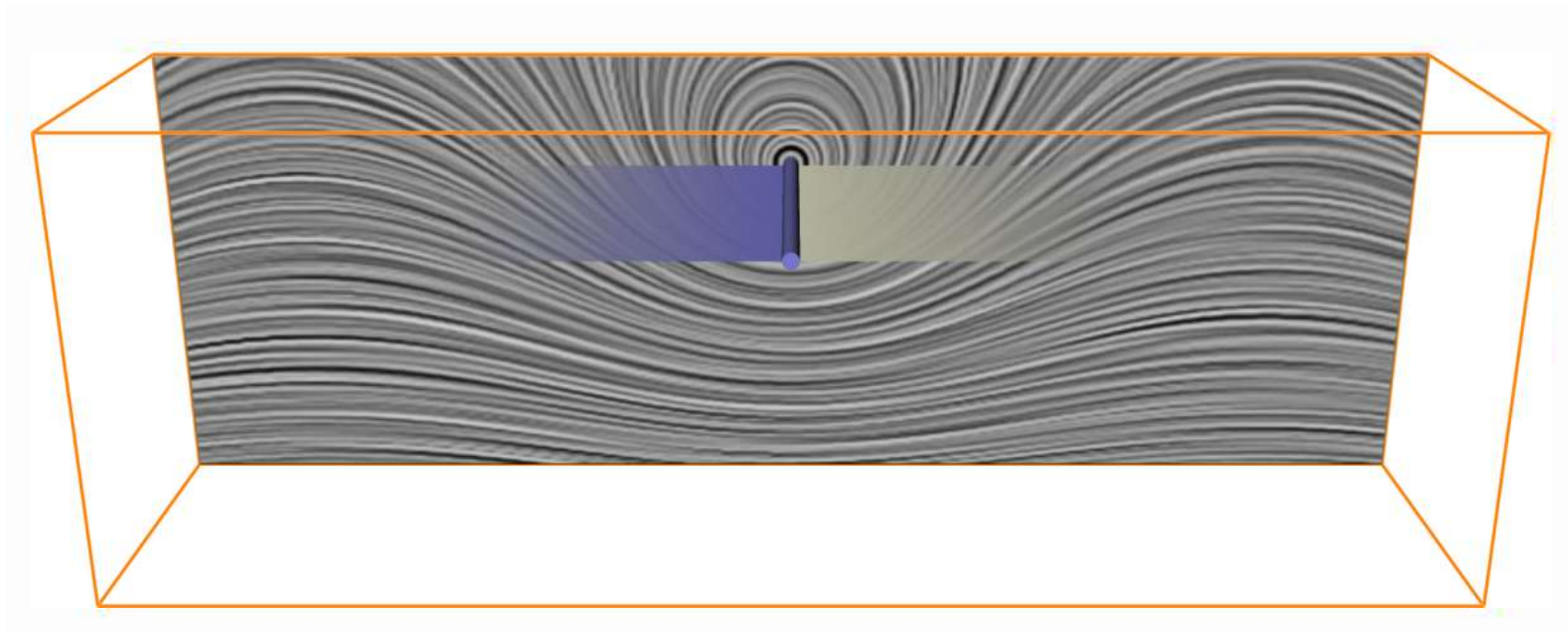


Sujudi/Haimes: Overview



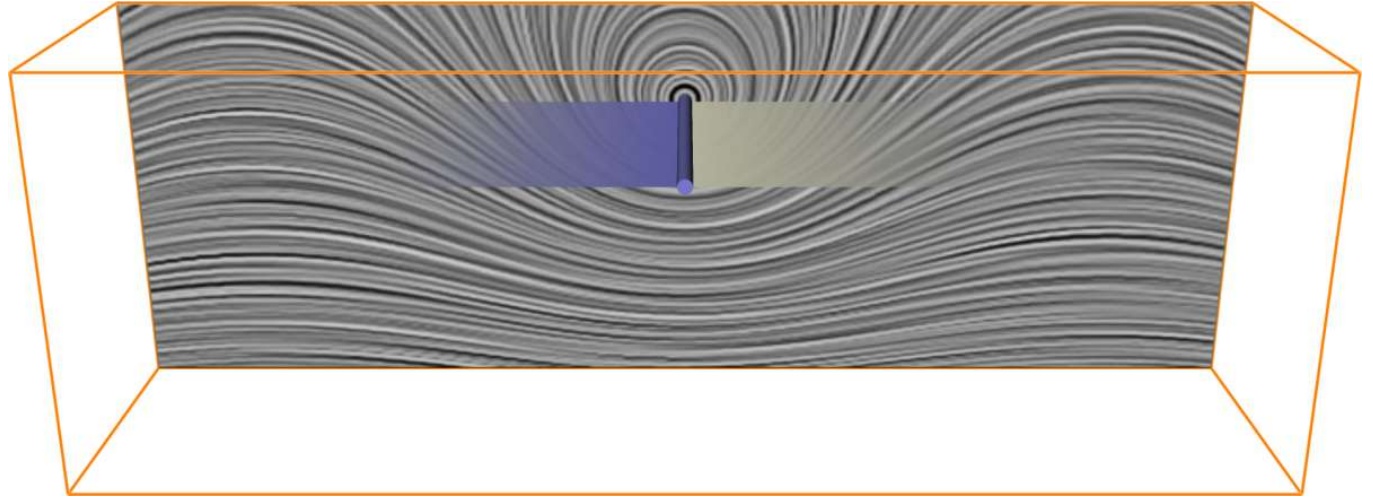
Swirling motion of stream lines – 3D unsteady

$$\mathbf{s} = \begin{pmatrix} u \\ v \\ w \\ 0 \end{pmatrix}$$

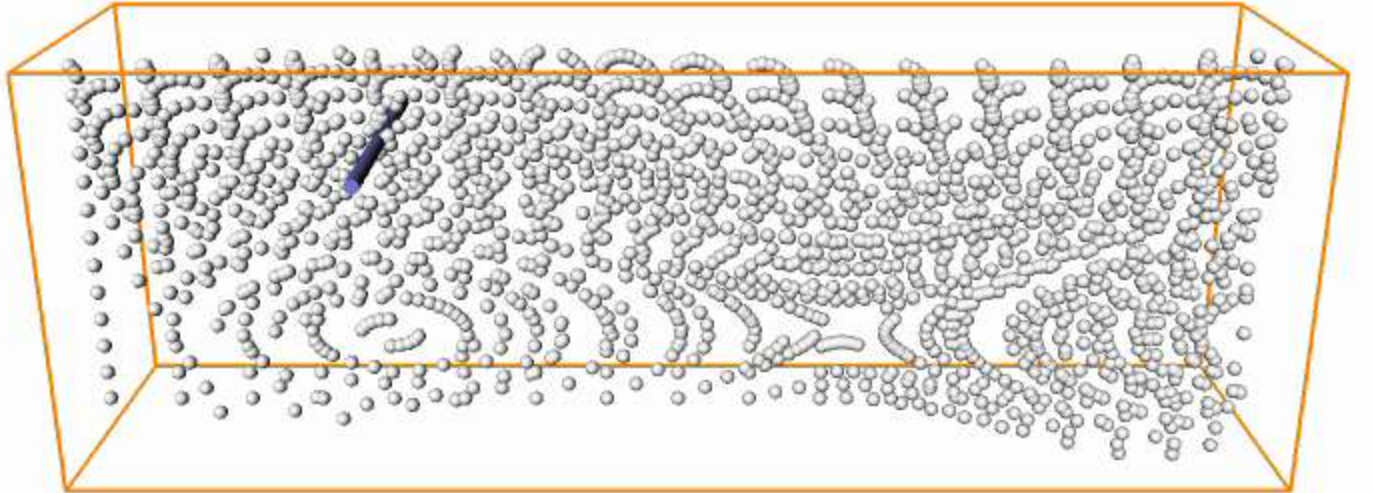


Swirling motion of path lines – 3D unsteady

$$\mathbf{s} = \begin{pmatrix} u \\ v \\ w \\ 0 \end{pmatrix} \quad \text{stream lines}$$



$$\mathbf{p} = \begin{pmatrix} u \\ v \\ w \\ 1 \end{pmatrix} \quad \begin{array}{l} \text{path lines} \\ \text{particles} \end{array}$$



Swirling motion of path lines – 3D unsteady

$$\mathbf{p} = \begin{pmatrix} u \\ v \\ w \\ 1 \end{pmatrix} \quad \begin{array}{l} \text{path lines} \\ \text{of a 3D unsteady flow} \end{array}$$

$$\mathbf{J}(\mathbf{p}) = \begin{bmatrix} u_x & u_y & u_z & u_t \\ v_x & v_y & v_z & v_t \\ w_x & w_y & w_z & w_t \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

→ eigenvalues

e_1	e_2	e_3	0
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complex (necessary condition)

two real eigenvectors $\mathbf{e}^s$ $\mathbf{f}$

$$\lambda_1 \mathbf{p} + \lambda_2 \mathbf{e}^s + \lambda_3 \mathbf{f} = \mathbf{0}$$

with $\sum \lambda_i^2 > 0$

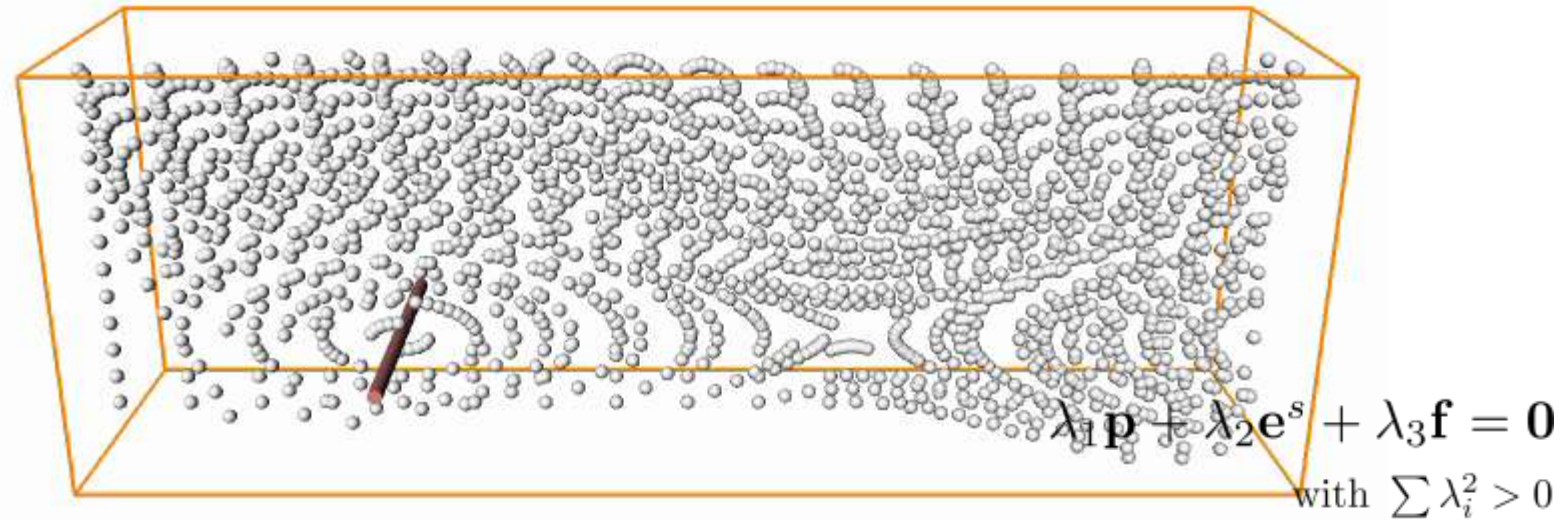
Coplanarity of
three 4D vectors



Original idea of Sujudi/Haimes:
Although a point $\mathbf{x}$ on the core structure is surrounded by spiraling integral curves, the flow vector at $\mathbf{x}$ itself is solely governed by the non-swirling part of the flow.

Swirling motion of path lines – 3D unsteady

$$\mathbf{p} = \begin{pmatrix} u \\ v \\ w \\ 1 \end{pmatrix} \quad \text{path lines} \\ \text{of a 3D unsteady flow} \quad \longrightarrow$$



Coplanar Vectors operator

Coplanar Vectors operator

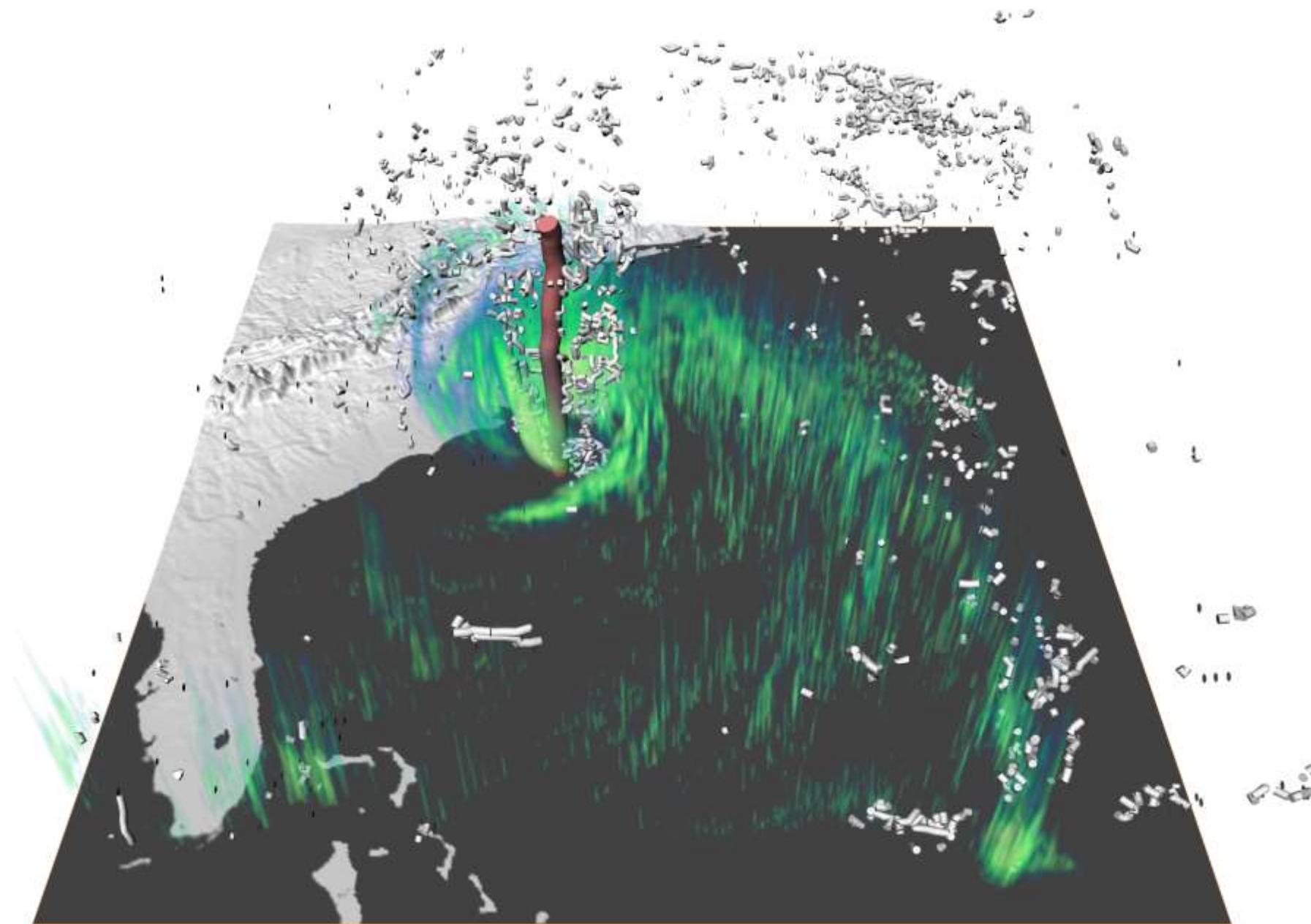
$$\lambda_1 \mathbf{p} + \lambda_2 \mathbf{e}^s + \lambda_3 \mathbf{f} = \mathbf{0}$$

$$\lambda_1 \begin{pmatrix} u \\ v \\ w \\ 1 \end{pmatrix} + \lambda_2 \begin{pmatrix} e_1^s \\ e_2^s \\ e_3^s \\ 0 \end{pmatrix} + \lambda_3 \begin{pmatrix} f_1 \\ f_2 \\ f_3 \\ f_4 \end{pmatrix} = \mathbf{0}$$

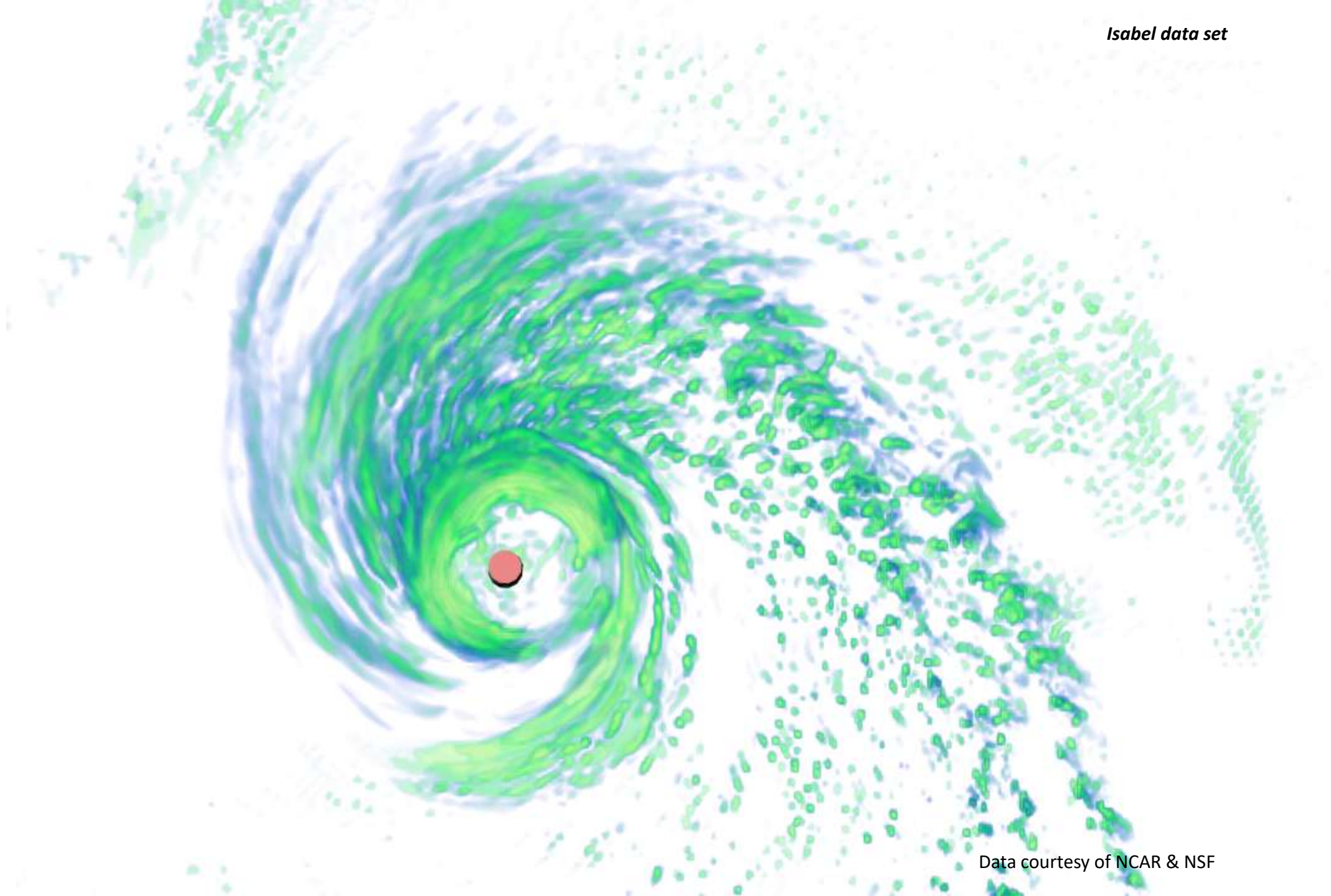
$\lambda_1 = -\lambda_3 f_4$

$$\underbrace{\lambda_2 \begin{pmatrix} e_1^s \\ e_2^s \\ e_3^s \end{pmatrix}}_{\mathbf{a}} + \underbrace{\lambda_3 \left(\begin{pmatrix} f_1 \\ f_2 \\ f_3 \end{pmatrix} - f_4 \begin{pmatrix} u \\ v \\ w \end{pmatrix} \right)}_{\mathbf{b}} = \mathbf{0}$$

Parallel Vectors operator



Data courtesy of NCAR & NSF



Cores of Swirling Particle Motion in Unsteady Flows
Weinkauff, Sahner, Theisel, Hege

Cavity Data Set

IEEE Visualization 2007

Summary: Vortex Core Lines

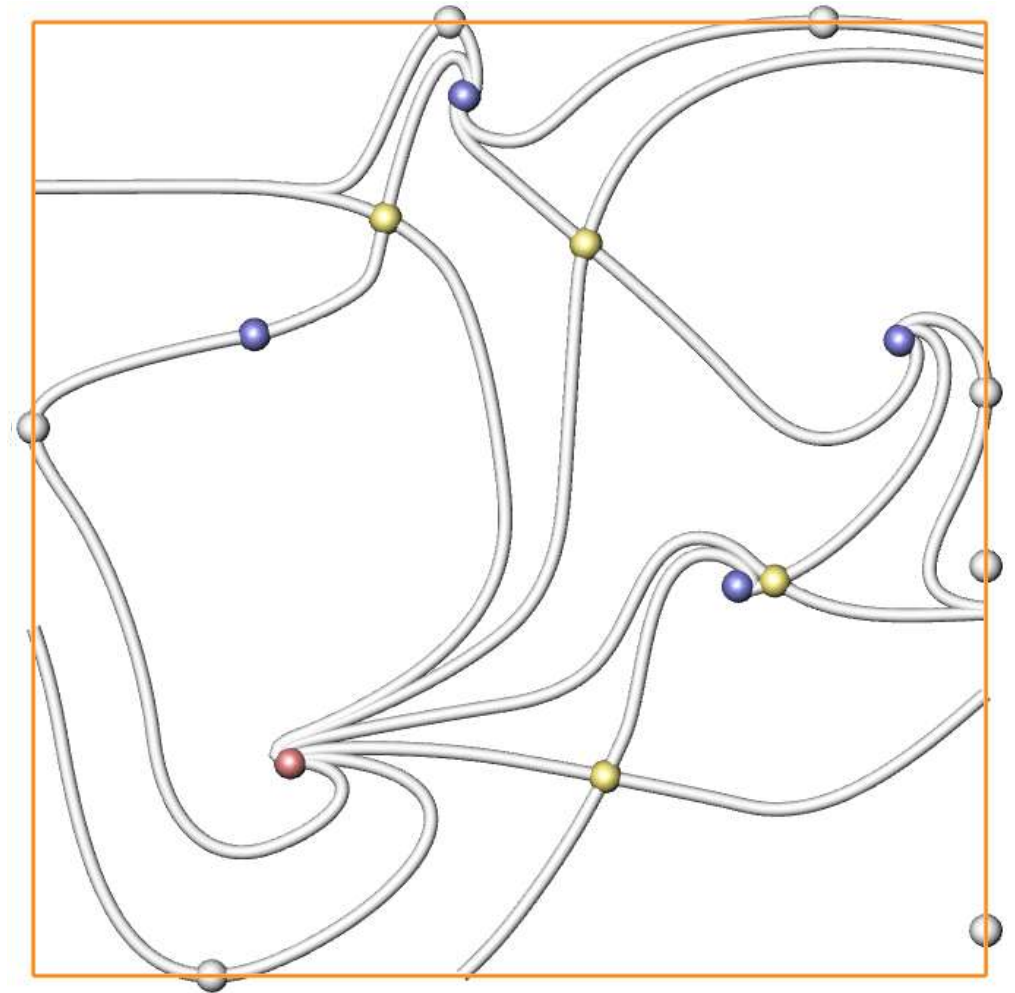
- A vortex is a region of swirling flow around a vortex core line
- Many mathematical formalizations, none of them universally accepted
- Vortex visualization
 - Region-based (isosurfaces, volume rendering)
 - Line-based (extraction of vortex core lines)
 - Prediction/correction
 - Parallel vectors operator
 - Coplanar vectors (unsteady case)

Section 13.2:

Vector Field Topology

Vector Field Topology

- Basic idea of vector field topology:
 - Segment domain into regions in which all streamlines have the same asymptotic behavior
 - Same starting point, same end point
 - Point and line features that form the “topological skeleton”



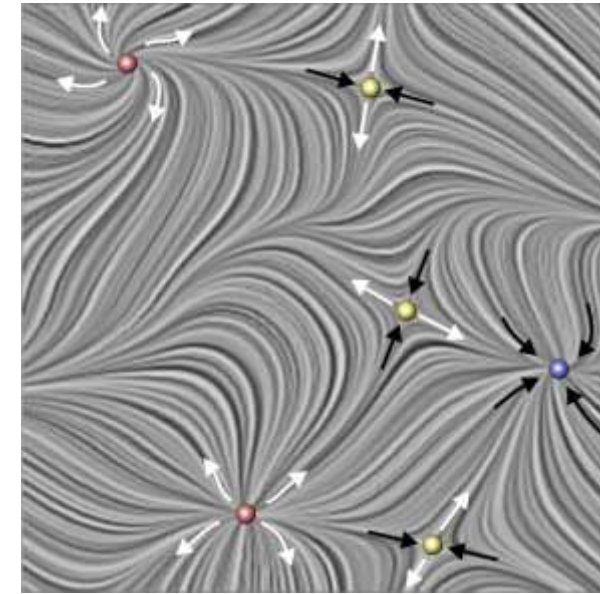
Topological Structures

steady 2D vector field

$$\mathbf{v} = \begin{pmatrix} u \\ v \end{pmatrix}$$

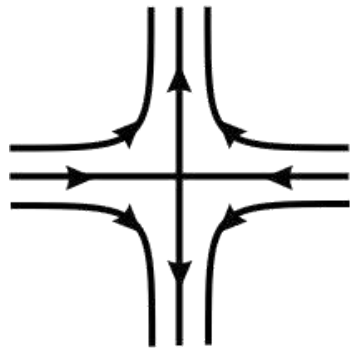


stream lines

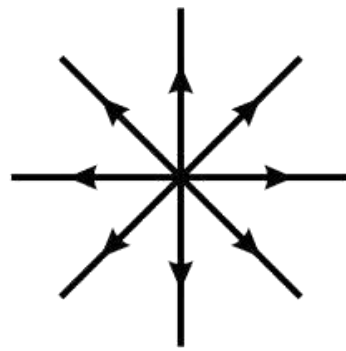


critical points

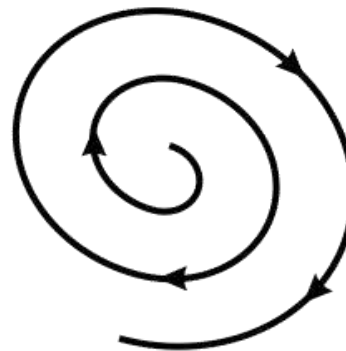
$$\mathbf{v}(\mathbf{x}_0) = \mathbf{0} \quad \text{with} \quad \mathbf{v}(\mathbf{x}_0 \pm \epsilon) \neq \mathbf{0}$$



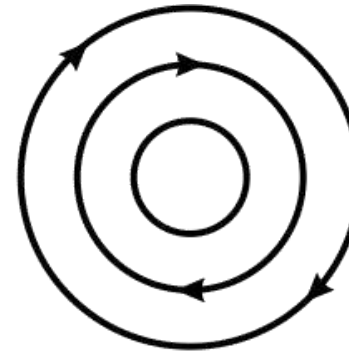
Saddle point
 $R_1 < 0, R_2 > 0$
 $I_1 = I_2 = 0$



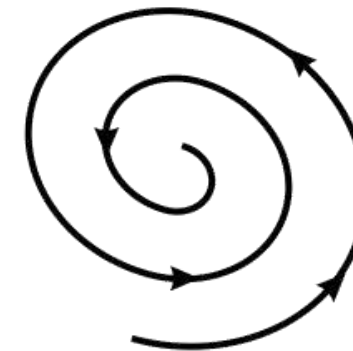
Repelling node
 $R_1, R_2 > 0$
 $I_1 = I_2 = 0$



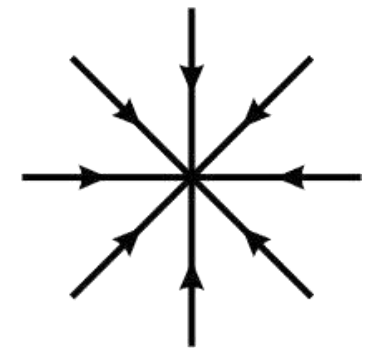
Repelling focus
 $R_1 = R_2 > 0$
 $I_1 = -I_2 \neq 0$



Center
 $R_1 = R_2 = 0$
 $I_1 = -I_2 \neq 0$



Attracting focus
 $R_1 = R_2 < 0$
 $I_1 = -I_2 \neq 0$



Attracting node
 $R_1, R_2 < 0$
 $I_1 = I_2 = 0$

First-order 2D Critical Points

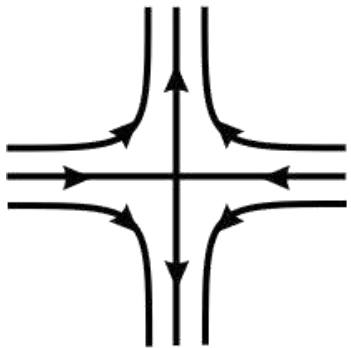
Definition: The point $\mathbf{x}_0$ is a first order critical point of the vector field $\mathbf{v}$ if

1) $\mathbf{x}_0$ is a critical point of $\mathbf{v}$, and

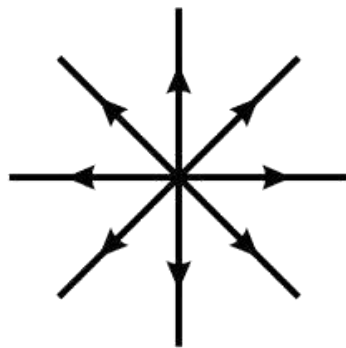
2) $\det(\mathbf{J}_{\mathbf{v}}(\mathbf{x}_0)) \neq 0$

– Classification by the eigenvalues of the Jacobian:

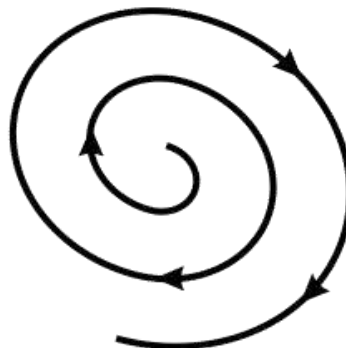
- $R_1, R_2 \rightarrow$ real part of the eigenvalue
 $I_1, I_2 \rightarrow$ imaginary part of the eigenvalue



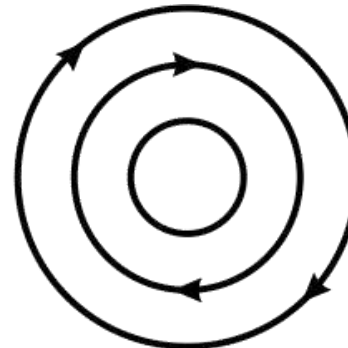
Saddle point
 $R_1 < 0, R_2 > 0$
 $I_1 = I_2 = 0$



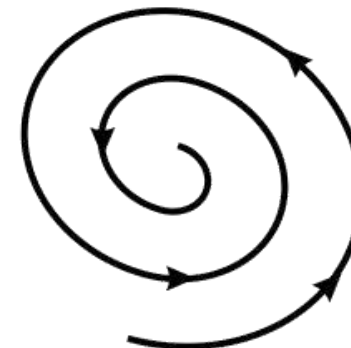
Repelling node
 $R_1, R_2 > 0$
 $I_1 = I_2 = 0$



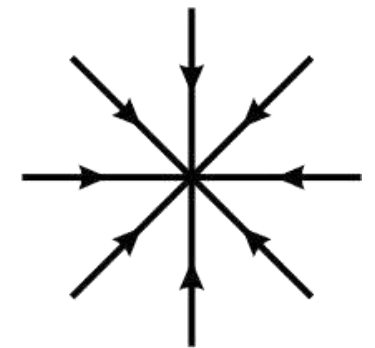
Repelling focus
 $R_1 = R_2 > 0$
 $I_1 = -I_2 \neq 0$



Center
 $R_1 = R_2 = 0$
 $I_1 = -I_2 \neq 0$



Attracting focus
 $R_1 = R_2 < 0$
 $I_1 = -I_2 \neq 0$

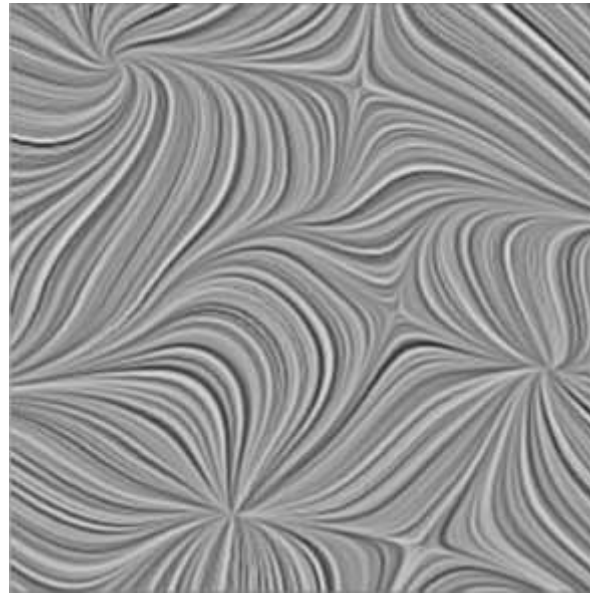


Attracting node
 $R_1, R_2 < 0$
 $I_1 = I_2 = 0$

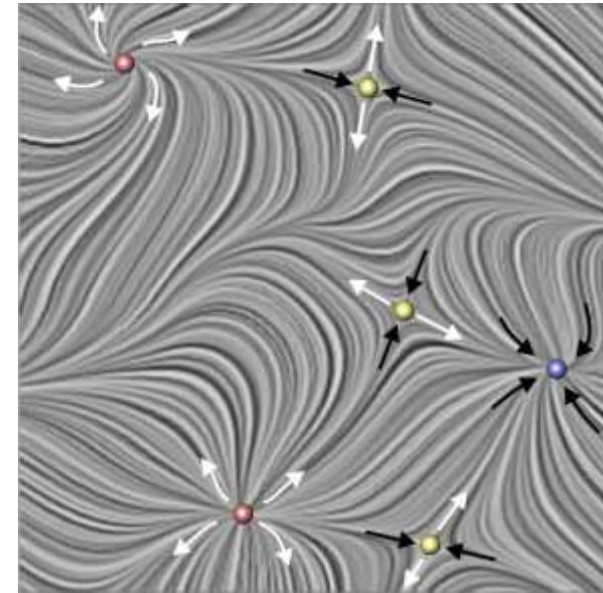
Topological Structures

steady 2D vector field

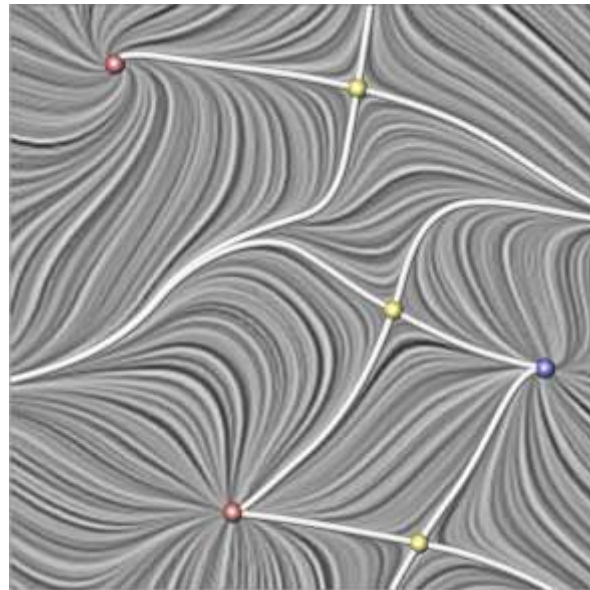
$$\mathbf{v} = \begin{pmatrix} u \\ v \end{pmatrix}$$



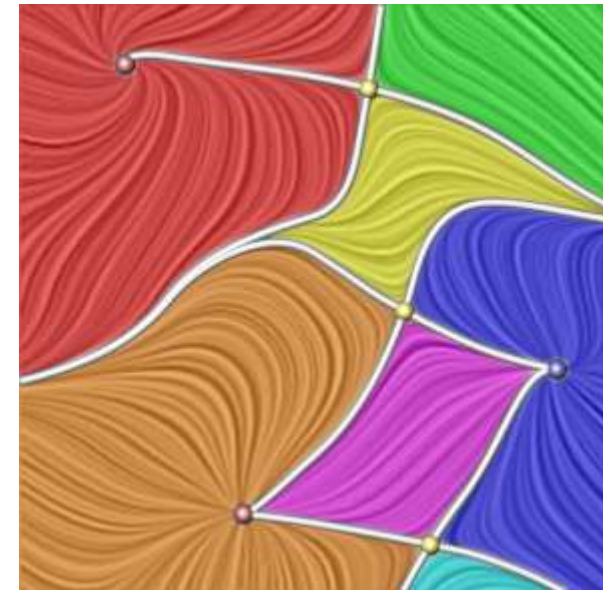
stream lines



critical points



separation lines



sectors of different flow behavior

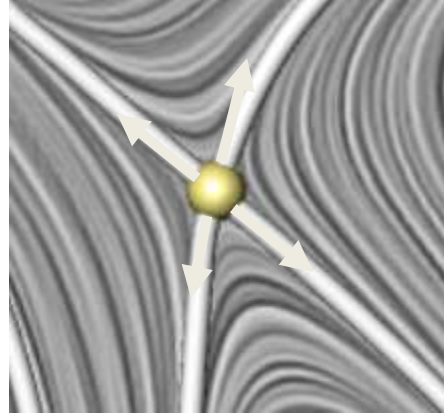
Separatrices

Definition: A separatrix is a stream line that separates regions of asymptotically different flow behavior.

- Separatrices are **global features**, we cannot decide if a point belongs to a separatrix just by looking at its immediate neighborhood!
- **Types of separatrices:**
 1. Each stream line originating/ending in a critical point and separating two sectors is a separatrix.
 2. Isolated closed stream lines
 3. Stream lines through boundary switch points

Separatrices From First Order Saddles

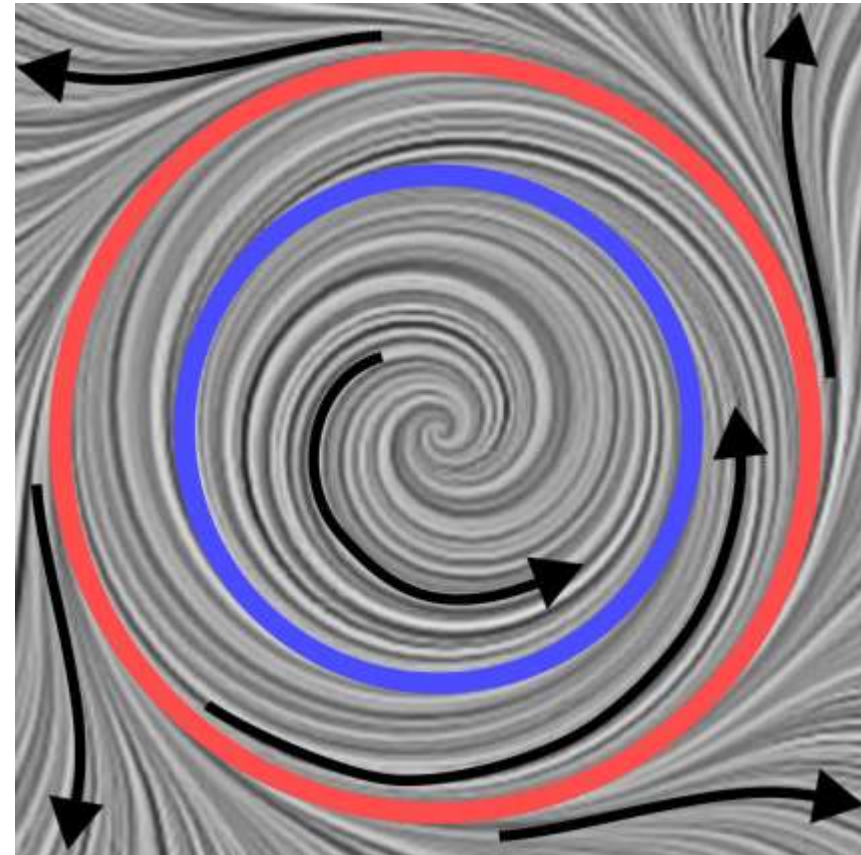
At each saddle, we need to start integration of four separatrices, in the directions of the two eigenvectors of the Jacobian matrix.



Closed Stream Lines

Localization of closed stream lines is outside our scope

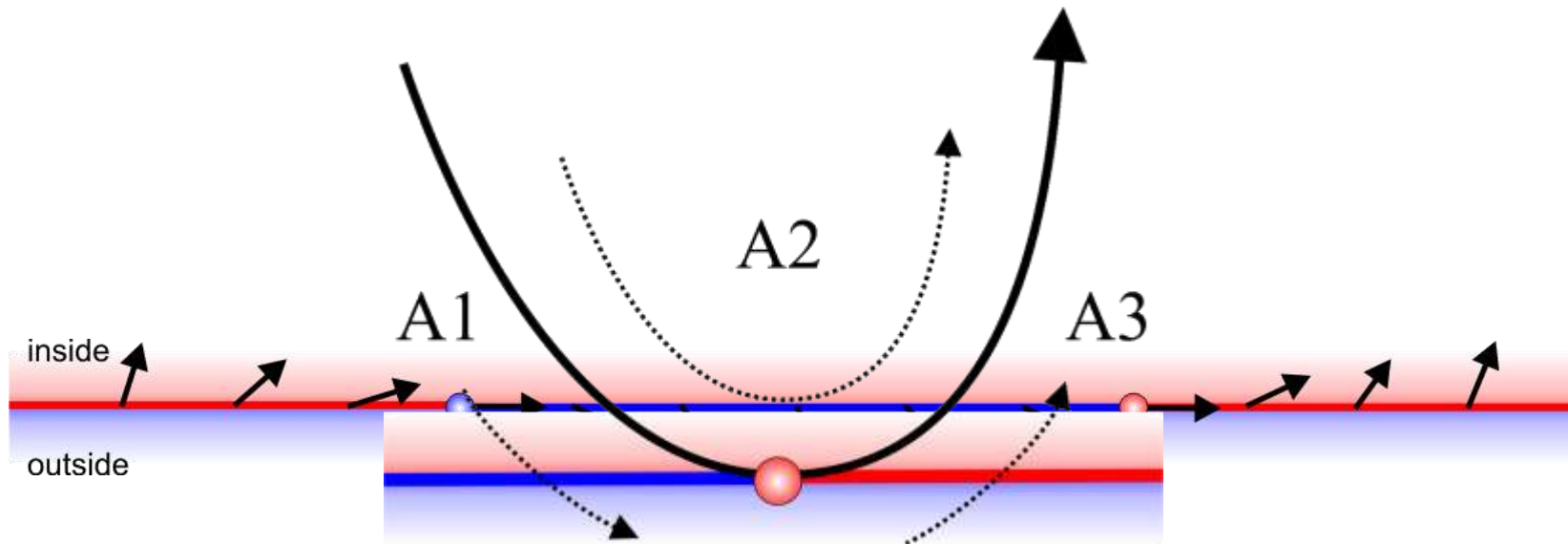
[Theisel et al 2003]:
Intersecting stream surfaces
in a derived 3D vector field



Boundary Switch Points

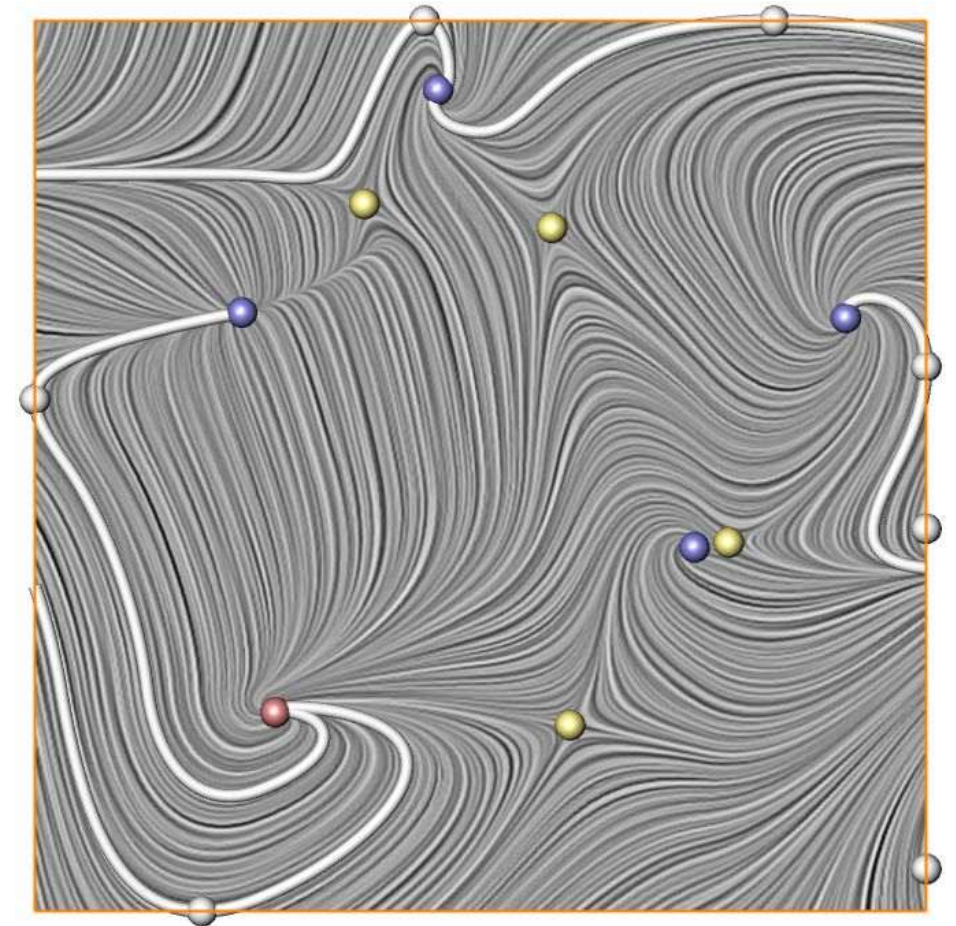
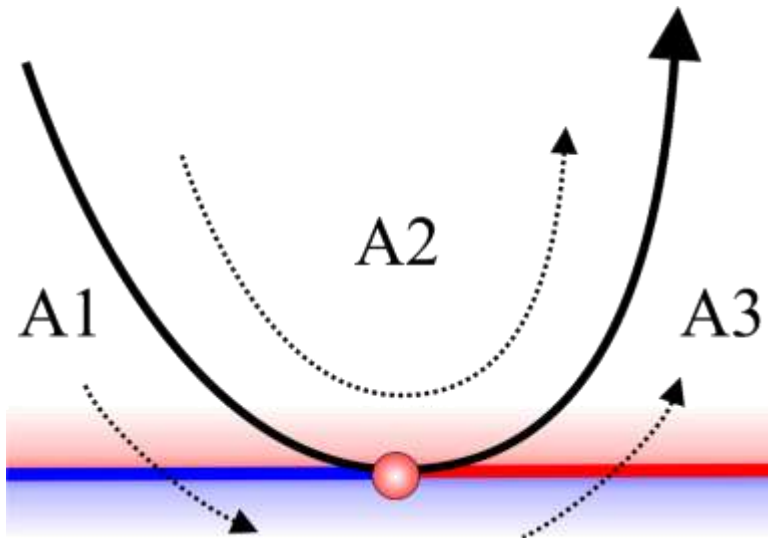
Definition: A point $\mathbf{x}$ on the boundary of the domain of a vector field $\mathbf{v}$ is a *boundary switch point* if $\mathbf{v}(\mathbf{x})$ is parallel to the tangent of the boundary curve at $\mathbf{x}$

- Switching between inflow $\Leftrightarrow$ outflow at the boundary

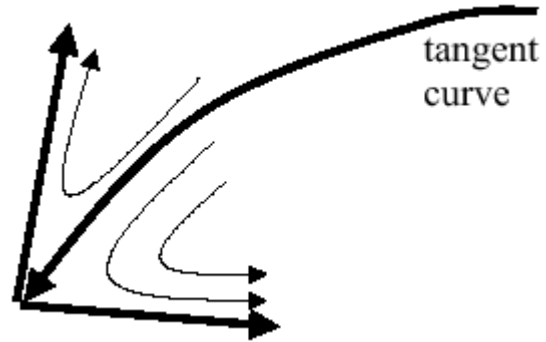


Separatrices from Boundary Switch Points

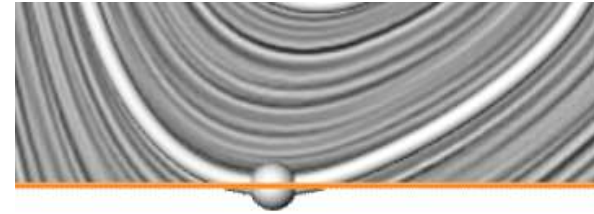
We need to start integrating additional separatrices at boundary switch points.



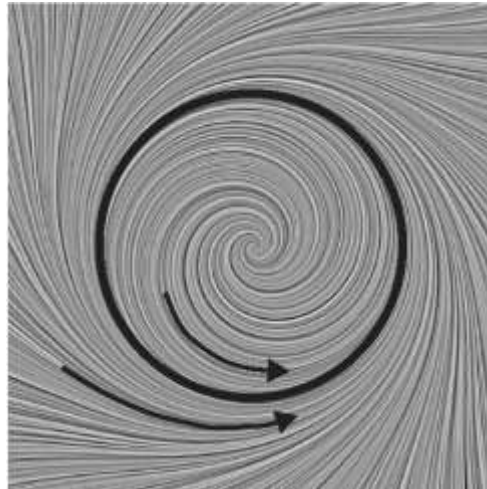
Summary: Three Types of Separatrices



Starting at Saddle Points

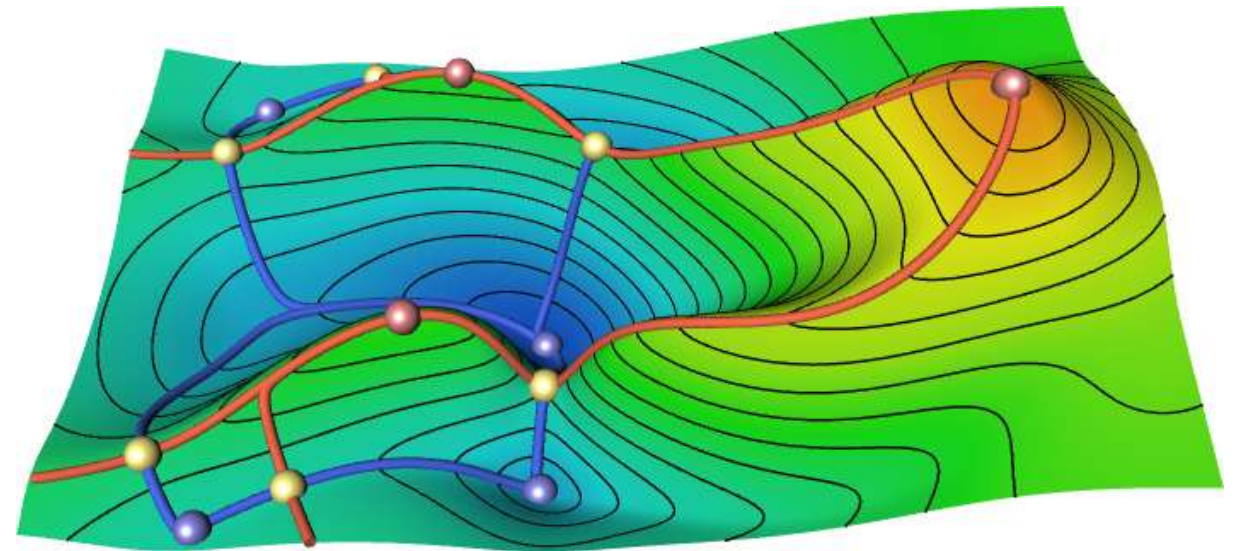
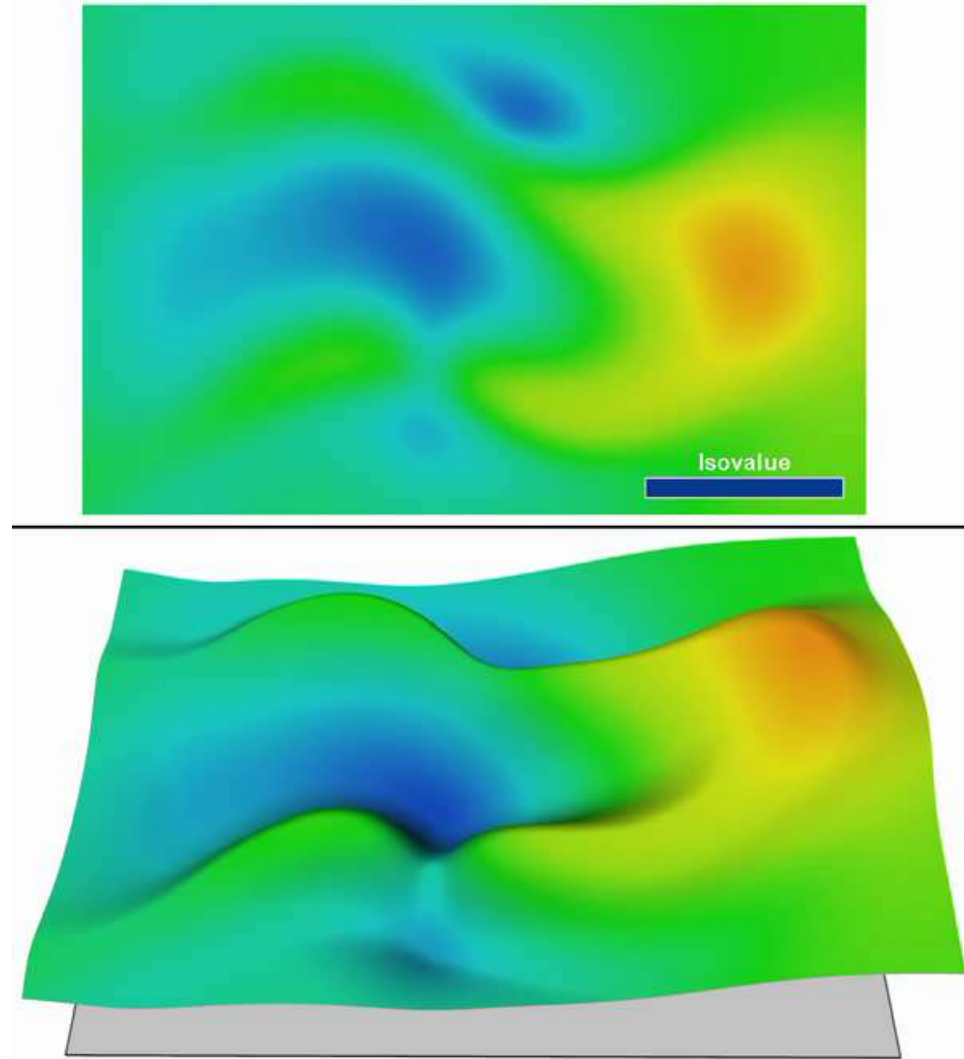


Boundary Switch Points



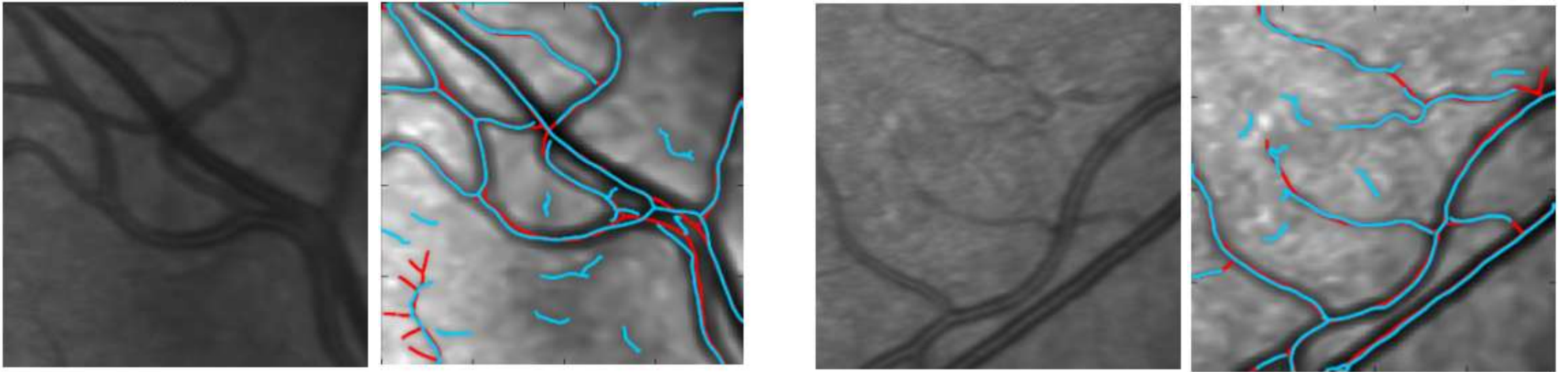
Closed Stream Lines

Intuition: Watershed Transformation



Example: Valley Lines for Vessel Extraction

- [Gorgi Zadeh et al. 2017]:
 - *Left*: Part of a fundus image
 - *Right*: Filtered; **vessel locations** and **valley curves**



Summary: Vector Field Topology

- **Topological skeleton**
 - Consists of critical points, boundary switch points, separatrices and closed stream lines
 - Partitions the domain of the vector field into regions of the same asymptotic behavior
 - Reduces the vector field to its “structurally relevant” information
- One practical application: **Watershed transformation**
 - Amounts to a subset of the topological skeleton of the gradient vector field

References on Vortex Detection

Overview:

- Chapter 14 of the Visualization Handbook: “Detection and Visualization of Vortices”, by M. Jiang, R. Machiraju, D. Thompson

Classical References:

- H. J. Lugt. Vortex Flow in Nature and Technology. Wiley, 1972.
- S. K. Robinson. Coherent Motions in the Turbulent Boundary Layer. Ann. Rev. Fluid Mechanics, 23:601–639, 1991.
- J. Jeong and F. Hussain. On the Identification of a Vortex. J. Fluid Mechanics, 285:69–94, 1995.
- L. M. Portela. Identification and Characterization of Vortices in the Turbulent Boundary Layer. PhD thesis, Stanford University, 1997.

References on Vortex Core Lines

- D. C. Banks and B. A. Singer. A Predictor-Corrector Technique for Visualizing Unsteady Flow. IEEE Trans. on Visualization and Computer Graphics, 1(2):151–163, 1995.
- M. Roth and R. Peikert. A Higher-Order Method for Finding Vortex Core Lines. IEEE Visualization 1998, pp. 143–150
- Jan Sahner, Tino Weinkauff, Hans-Christian Hege: Galilean Invariant Extraction and Iconic Representation of Vortex Core Lines. EuroVis 2005, pp. 151-160

References on Vector Field Topology

- Chapter 17 of the Visualization Handbook: “Topological Methods for Flow Visualization”, by Gerik Scheuermann and Xavier Tricoche
- J.L. Helman, L. Hesselink: Visualizing Vector Field Topology in Fluid Flows. IEEE Computer Graphics and Applications 11(3):36-46, 1991
- Xavier Tricoche: Vector and Tensor Field Topology Simplification, Tracking, and Visualization. PhD Thesis, Universität Kaiserslautern, 2002
- Tino Weinkauff: Extraction of Topological Structures in 2D and 3D Vector Fields. PhD Thesis, Otto-von-Guericke-Universität Magdeburg, 2008

Exam-like Questions

1. State an advantage of feature-based visualization, and name an example of a feature-based visualization technique.
2. Part of the Sujudi/Haimes criterion for vortex core lines is that the local vector is an eigenvector of the vector field Jacobian. What additional condition needs to be met?
3. What is the role of a separatrix in the topological skeleton of a steady 2D vector field?